

1. Choose a region. Determine which region on AWS to deploy EC2 on. Regions in AWS are effectively geographic locations of services.
2. Create a virtual private cloud (VPC) and subnet, which is a logically isolated set of services in AWS.
3. Create key pairs and a security group, which are needed to access the virtual machines or containers in AWS.
4. Explain process of deploying an EC2 instance
5. Implement the deployment in the way you explained it.

Note: create an IAM user for admin access if you don't have one, then use it to perform the tasks.

Step 1: Choosing a region.

Edit localization and default Region [Info](#)
Manage localization and default Region settings for the current user. These settings do not apply to other IAM users or accounts.

Localization
Language
English (US) ▼

Default Region
Default Region
Choose the Region that will be selected by default in the AWS Console each time you log in.
United States (N. Virginia) us-east-1 ▼
Only Regions enabled for your account appear.

Cancel **Save settings**

United States (N. Virginia) ▼

Step 2. Create a VPC and Subnet

VPC:

aws

Search

[Alt+S]

United States (N. Virginia) ▼

Cindy ▼

VPC dashboard <

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Your VPCs (1) [Info](#)

Find VPCs by attribute or tag

Last updated less than a minute ago

Actions

Create VPC

☐	Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR
☐	-	vpc-0e9558a0a7ac95cb3	Available	Off	172.31.0.0/16	-

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

my-vpc

IPv4 CIDR block [Info](#)

- ☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

10.0.0.0/16

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

- ☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers [New](#)

Security

You successfully created vpc-05d16366c1eaca798 / my-vpc

vpc-05d16366c1eaca798 / my-vpc [Actions](#)

Details [Info](#)

VPC ID vpc-05d16366c1eaca798	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-06994f572ac7689c4	Main route table rtb-02e71aee99ff8c8d7
Main network ACL acl-079c7b1f6a7c3e4cc	Default VPC No	IPv4 CIDR 10.0.0.0/16	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 092896355827

[Resource map](#) | [CIDRs](#) | [Flow logs](#) | [Tags](#) | [Integrations](#)

Resource map [Info](#)

Subnet:

aws [Search] [Alt+S] United States (N. Virginia) Cindy

VPC > Your VPCs > vpc-05d16366c1eaca798

vpc-05d16366c1eaca798 / my-vpc

VPC dashboard < EC2 Global View [x] Filter by VPC

▼ **Virtual private cloud**

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers [New](#)

▼ **Security**

- Network ACLs
- Security groups

Details [Info](#)

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Resource map | CIDRs | Flow logs | Tags | Integrations

Resource map [Info](#)

VPC [Show details](#)

Your AWS virtual network

my-vpc

Subnets (1)

Subnets within this VPC

us-east-1a

my-subnet

Route tables (1)

Route network traffic to resources

rtb-02e71aeea9ff8c8d7

Network ACLs

Connect...

3. Create key pairs and a security group

Key Pair:

Create key pair [Info](#)

Key pair

A key pair, consisting of a private key and a public key, is a set of security credentials that you use to prove your identity when connecting to an instance.

Name

mykeypair1

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type [Info](#)

☒ RSA ☐ ED25519

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

Tags - optional

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#) [Create key pair](#)

aws [Search] [Alt+S] Cindy

EC2 > Key pairs

AMI Catalog

▼ **Elastic Block Store**

- Volumes
- Snapshots
- Lifecycle Manager

▼ **Network & Security**

- Security Groups
- Elastic IPs
- Placement Groups
- [Key Pairs](#)

Successfully created key pair

Key pairs (1/2) [Info](#)

Find Key Pair by attribute or tag

	Name	Type	Created	Fingerprint	ID
☒	mykeypair	rsa	2025/05/19 20:42 GMT-5	80:bb:bb:afa:3:d7:73:3b:f4:21:82:07:2d:4d:ff:...	key-07136...
☐	mykeypair1	rsa	2025/05/19 20:44 GMT-5	8a:10:7d:3f:01:69:e7:62:2c:a2:6b:81:dd:8a:e5:...	key-0ef6b...

mykeypair1.pem 1,678 B • Done

mykeypair.pem 1,678 B • 2 minutes ago

1252051573 GROUND - FedEx-Shipping-Label.pdf 36.3 KB • 7 hours ago

[Create key pair](#)

Security Group:

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

Inbound rules [Info](#)

Type [Info](#)

Protocol [Info](#)

Port range [Info](#)

Source [Info](#)

Description - optional [Info](#)

[Delete](#)[Add rule](#)

Outbound rules [Info](#)

Type [Info](#)

Protocol [Info](#)

Port range [Info](#)

Destination [Info](#)

Description - optional [Info](#)

[Delete](#)[Add rule](#)

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs. No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags

[Cancel](#)[Create security group](#)

EC2 > Security Groups > sg-0e6b15cb84395675d - allowssh

AMI Catalog

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

▼ Load Balancing

Load Balancers

Target Groups

Trust Stores

▼ Auto Scaling

Auto Scaling Groups

✔ Security group (sg-0e6b15cb84395675d | allowssh) was created successfully

Details

sg-0e6b15cb84395675d - allowssh

Actions

Details

Security group name

allowssh

Security group ID

sg-0e6b15cb84395675d

Description

Allow SSH from anywhere

VPC ID

vpc-05d16366c1eaca798

Owner

092896355827

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (1)

Search

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range
☐	-	sgr-074db106ee47881a0	IPv4	SSH	TCP	22

4. Explain process of deploying an EC2 instance

First, I chose the region location that is located closest to me and also based on latency, compliance, and location of user. I chose the region location by clicking on the top right of the AWS console and I chose the region (N. Virginia).

Second, I created a VPC. A VPC (Virtual Private Cloud) is a private network in AWS where resources can be launched. So I went to the VPC dashboard and created a VPC using IPV4 CIDR block 10.0.0.0/16. Next, I created the Subnet inside the VPC using 10.0.1.0/24. I made it public by later attaching an Internet Gateway and routing traffic through it.

On the third step, I created key pairs and security group. The key pair is used for SSH to access the instance. I downloaded the .pem file and stored it securely. The security group acts like a firewall. I created a security group with the rule allowing SSH (TCP port 22) from 0.0.0.0/0 (IP address).

Next, I went to the EC2 launch wizard. I chose and selected the AMI (Amazon Linux), I selected the instance type (T2), I attached the key pair I had created, and then I set the network settings by selecting the VPC and subnet that I had created earlier. I then enabled auto-assign public IP and chose the security group that I created earlier. I kept the default EBS storage, and finally launched the instance.

5. Implement the EC2 deployment in the way you explained it.

EC2 > Instances > Launch an instance

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

MyEC2

Add additional tags

▼ Application and OS Images (Amazon Machine Image)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2 AMI (HVM) – Kernel 5.10, SSD Volume Type
ami-0e58b56aa4d64231b (64-bit (x86)) / ami-0972357ba34d1ec40 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2 comes with five years support. It provides Linux kernel 5.10 tuned for optimal performance on Amazon EC2, systemd 219, GCC 7.3, Glibc 2.26, Binutils 2.29.1, and the latest software packages through extras. This AMI is the successor of the Amazon Linux AMI that is now under maintenance only mode and has been removed from this wizard.

Amazon Linux 2 Kernel 5.10 AMI 2.0.20250512.0 x86_64 HVM gp2

Architecture

AMI ID

Publish Date

Username

64-bit (x86)

ami-0e58b56aa4d64231b

2025-05-10

ec2-user

Verified provider

▼ Instance type Info | Get advice

Instance type

t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour
On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

mykeypair1

Create new key pair

▼ Network settings Info

VPC - required Info

vpc-05d16366c1eaca798 (my-vpc)
10.0.0.0/16

Subnet Info

subnet-0838b55ed92740442 my-subnet
VPC: vpc-05d16366c1eaca798 Owner: 092896355827 Availability Zone: us-east-1a
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.1.0/24

Auto-assign public IP Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

☐ Create security group

☒ Select existing security group

Common security groups [Info](#)

Select security groups

allowssh sg-0e6b15cb84395675d ✕
VPC: vpc-05d16366c1eaca798

 [Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► **Advanced network configuration**

▼ **Configure storage** [Info](#)

[Advanced](#)

1x GiB ▼ Root volume, Not encrypted

 Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage ✕

[Add new volume](#)

 Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies. 

► Summary

[Cancel](#)

[Launch instance](#)

 [Preview code](#)

 **Success**

Successfully initiated launch of instance ([i-07fe985e8945477f7](#))

► **Launch log**

EC2 > Instances

EC2

Dashboard

EC2 Global View

Events

Instances

- Instances
- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations

Images

- AMIs
- AMI Catalog

Elastic Block Store

- Volumes
- Snapshots

Instances (1/1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

☒	Name	Environment	Instance ID	Instance state	Instance type	Status checks
☒	MyEC2		i-07fe985e8945477f7	Running	t2.micro	2/2 checks passed

i-07fe985e8945477f7 (MyEC2)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance summary Info

Instance ID

i-07fe985e8945477f7

IPv6 address

—

Hostname type

IP name: ip-10-0-1-198.ec2.internal

Public IPv4 address

44.223.80.123 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-0-1-198.ec2.internal

Private IPv4 addresses

10.0.1.198

Public IPv4 DNS

—